

The Great Demotions

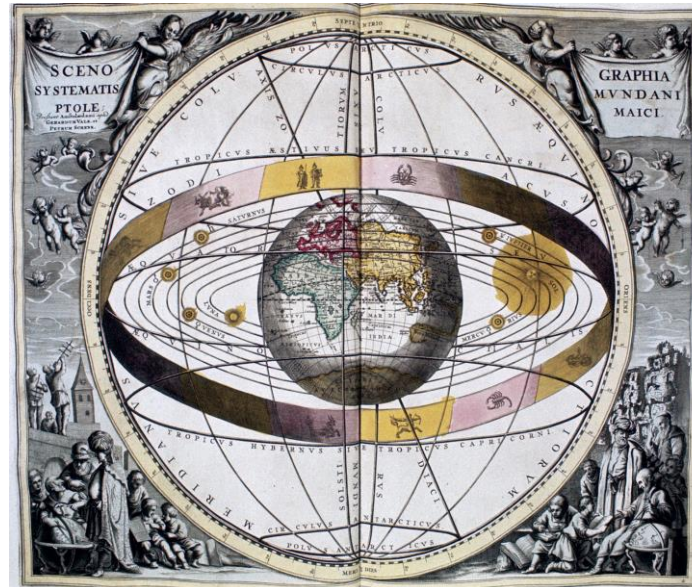
(revisited)



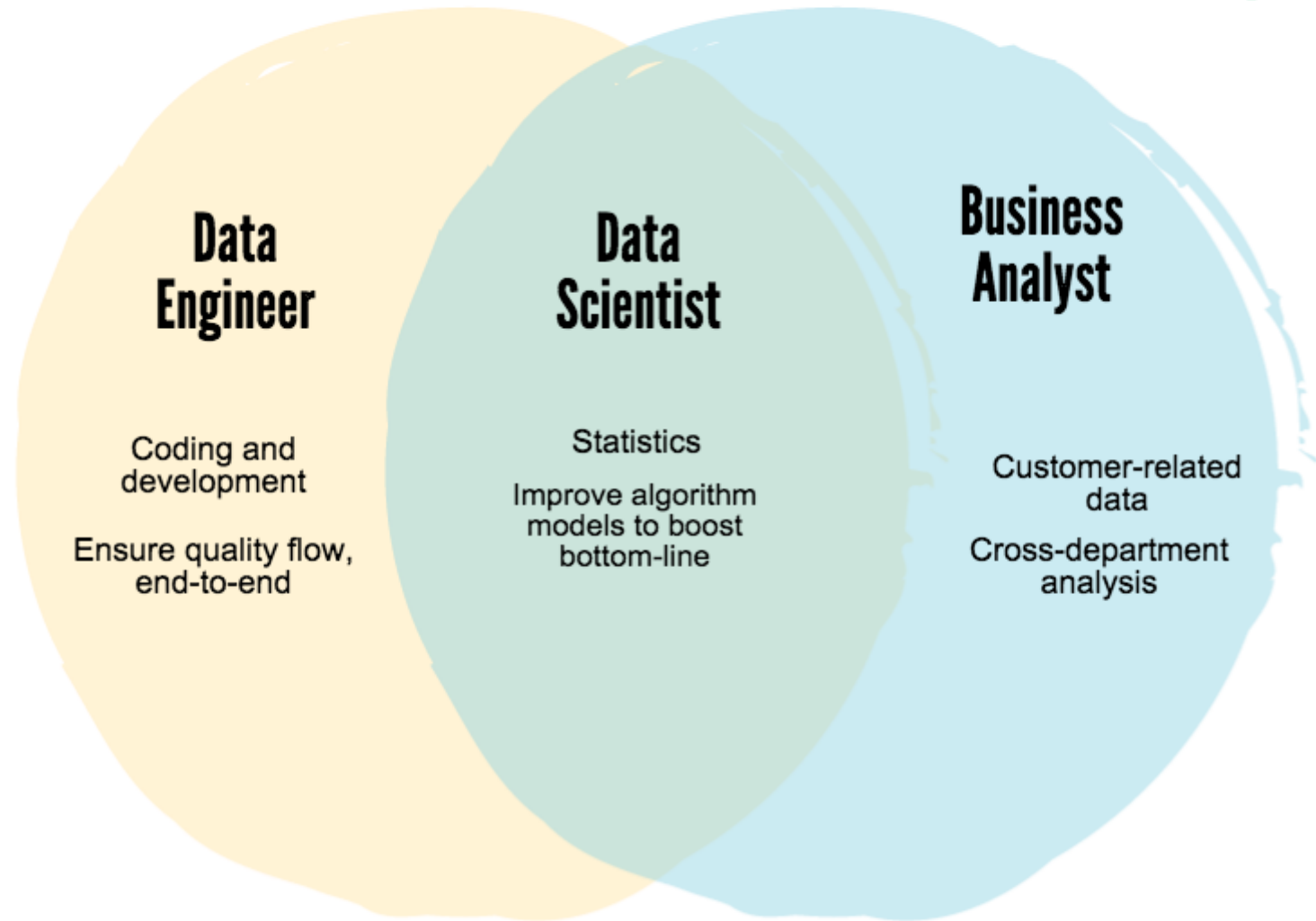
“you just can't differentiate between a robot and the very best of humans”
I Robot, Isaak Asimov 1953

[One philosopher] asserted that he knew the whole secret ... [H]e surveyed the two celestial strangers from top to toe, and maintained to their faces that their persons, their worlds, their suns, and their stars, were created solely for the use of man. At this assertion our two travelers let themselves fall against each other, seized with a fit of ... inextinguishable laughter.

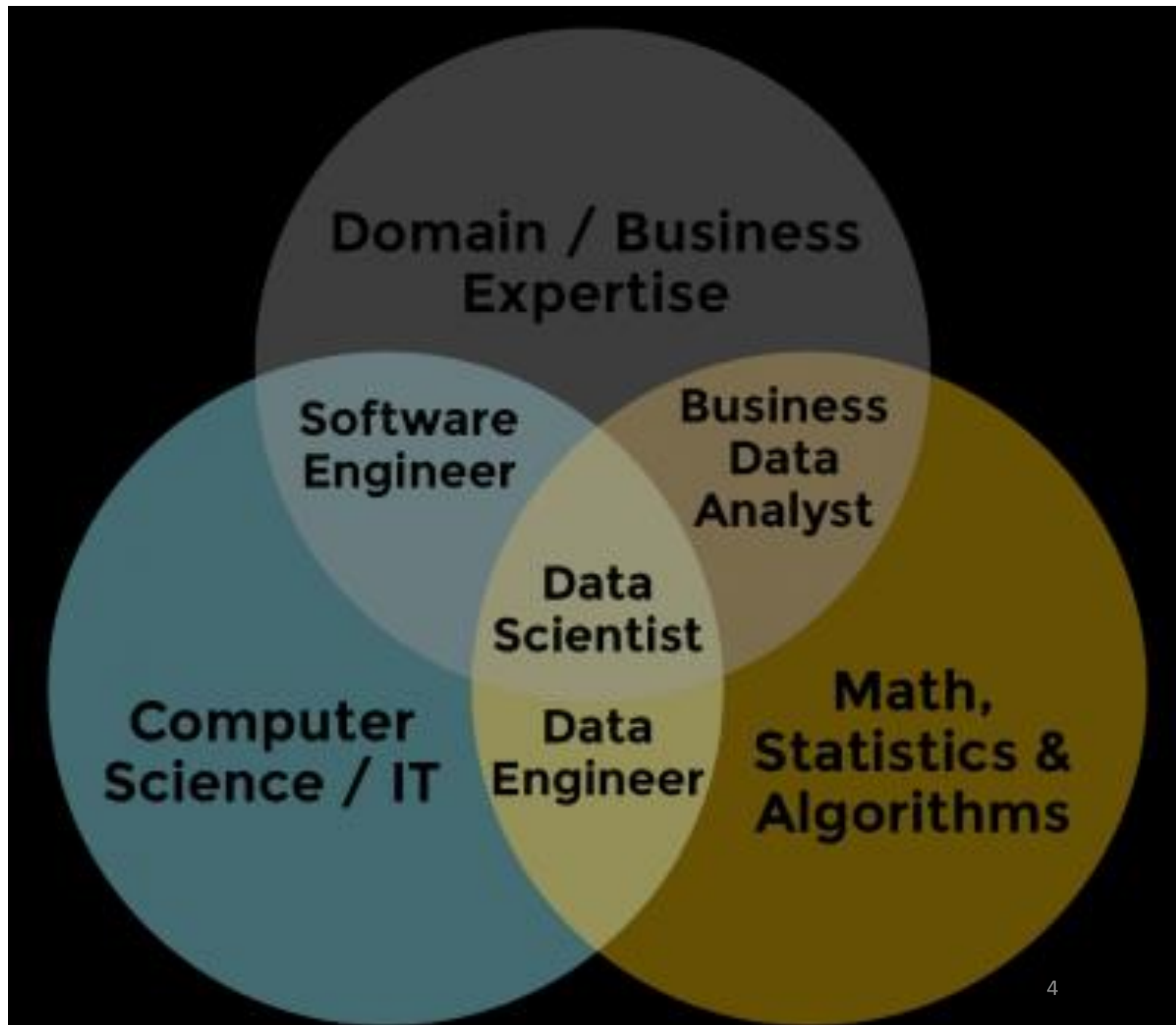
—VOLTAIRE, MICROMEGAS. A PHILOSOPHICAL HISTORY



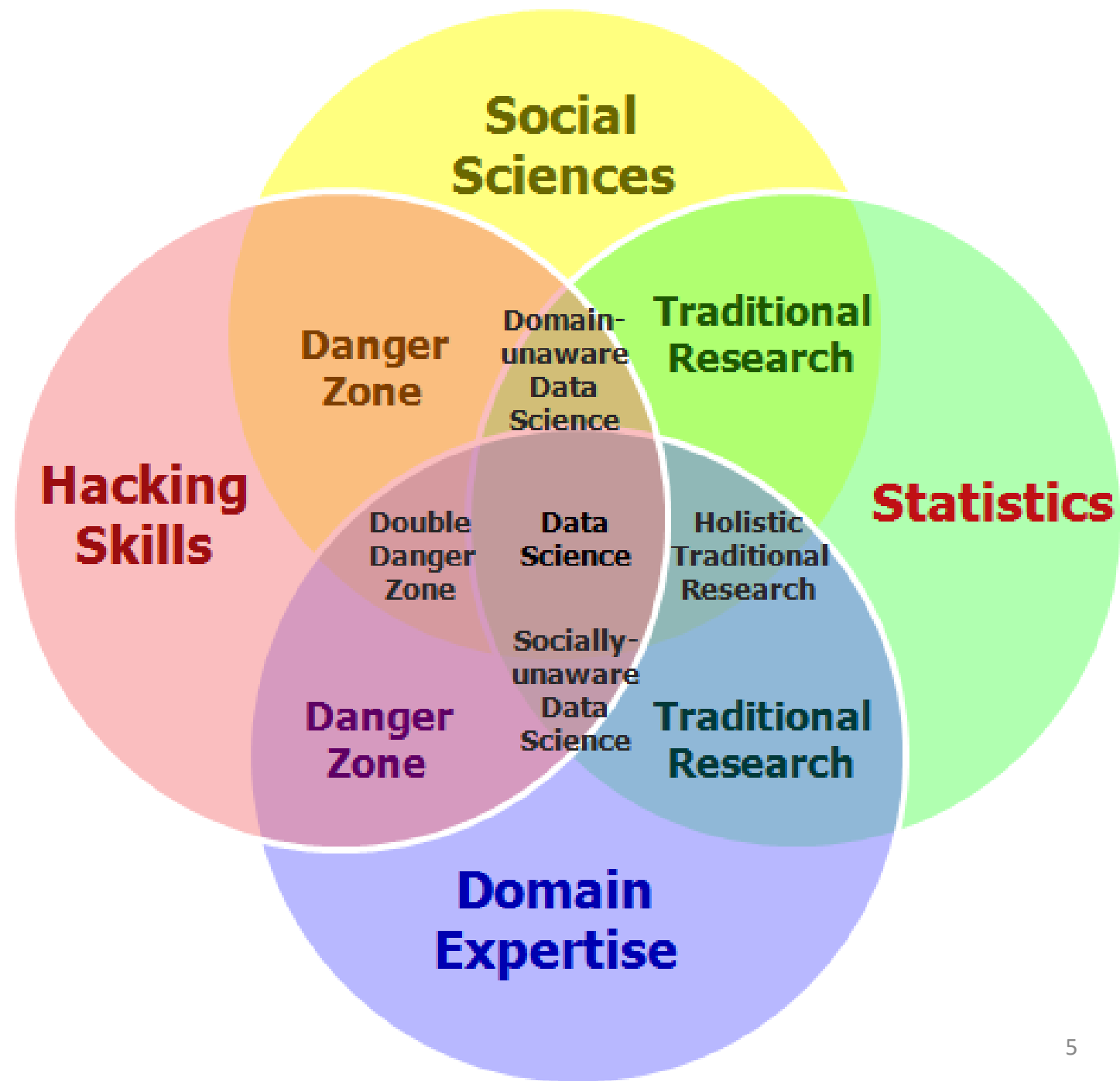
What is Data Science?



Right, but what
about Computer
Science?

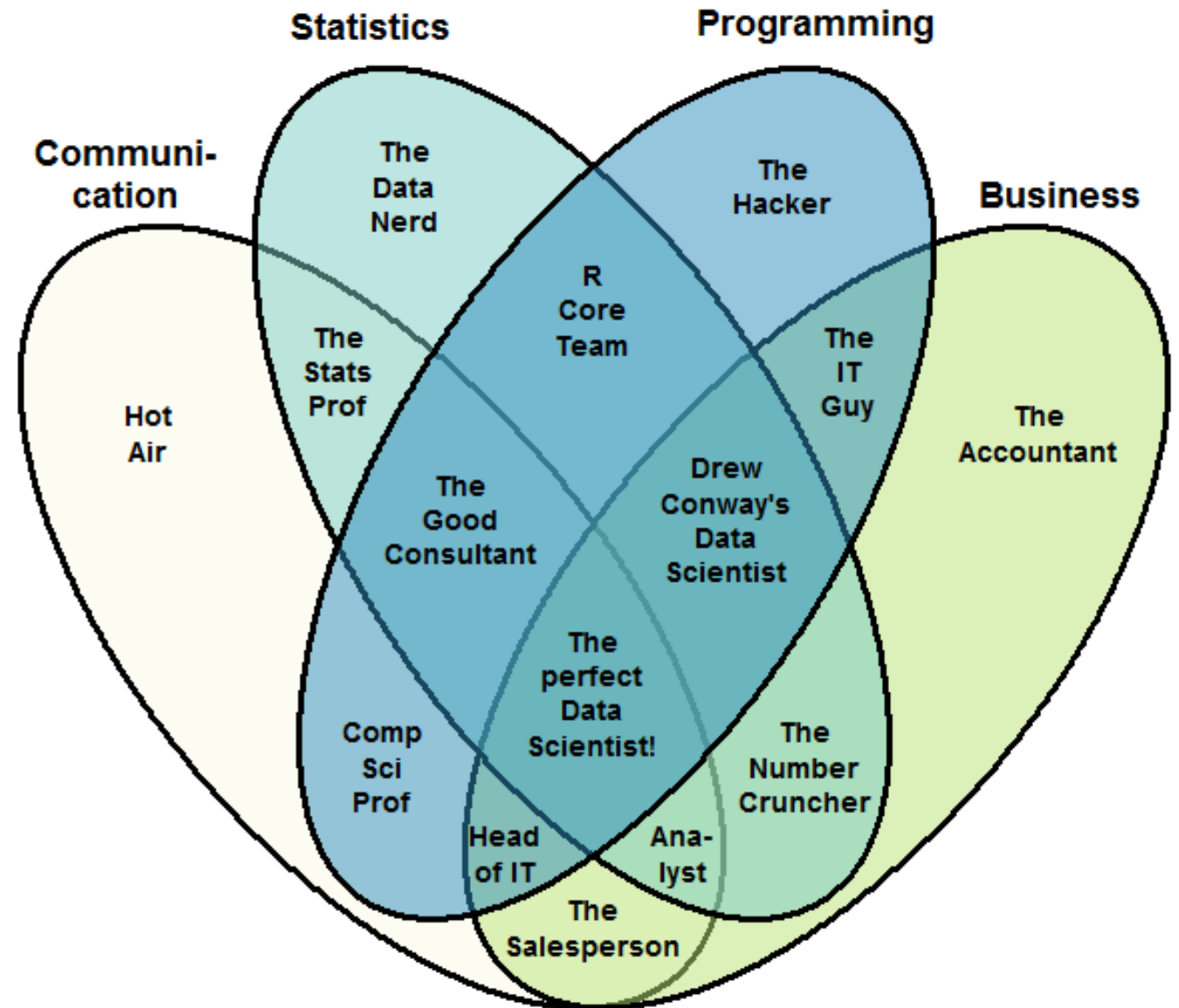


Surely More Circles
are better...

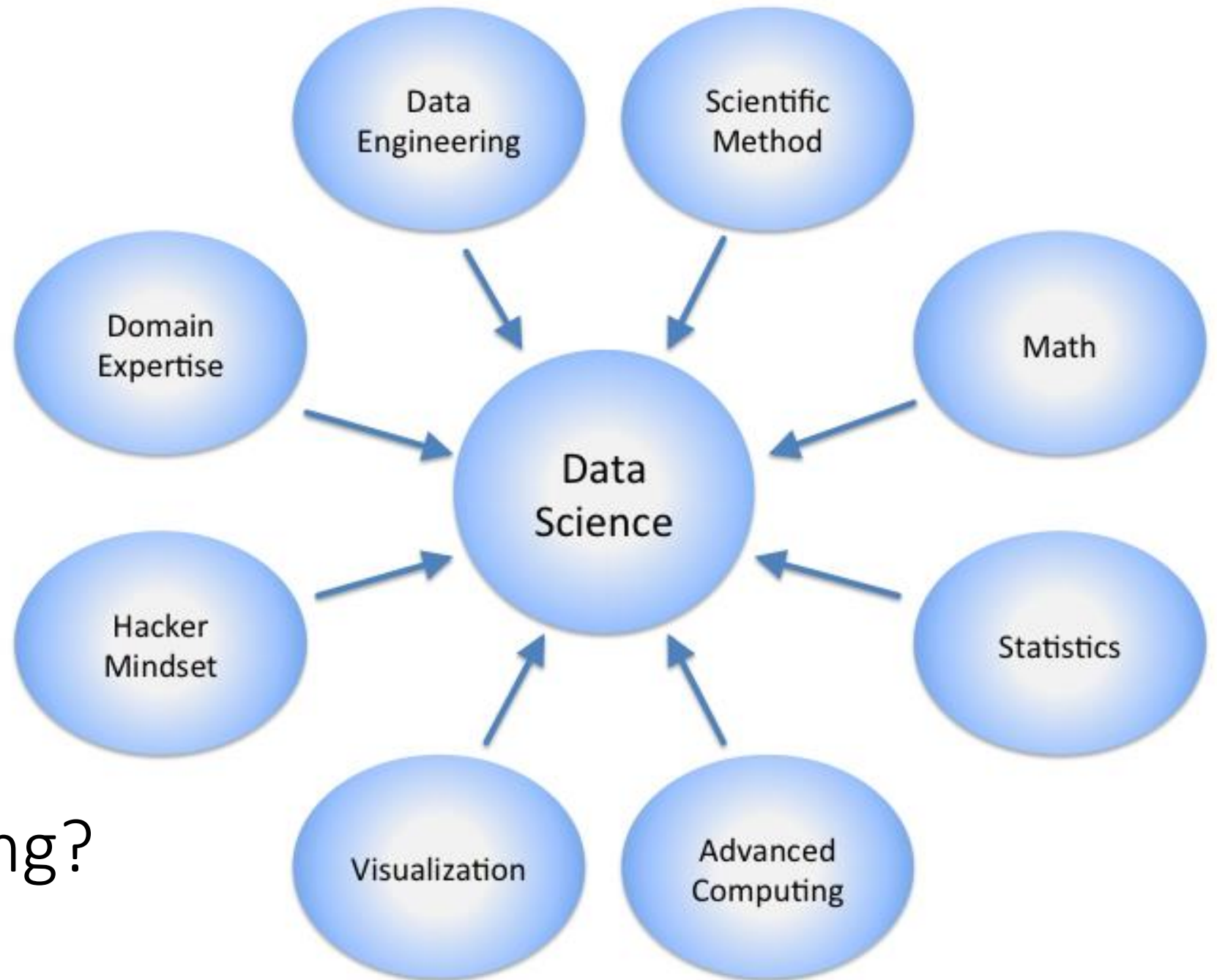


Maybe move away from circles altogether.

The Data Scientist Venn Diagram

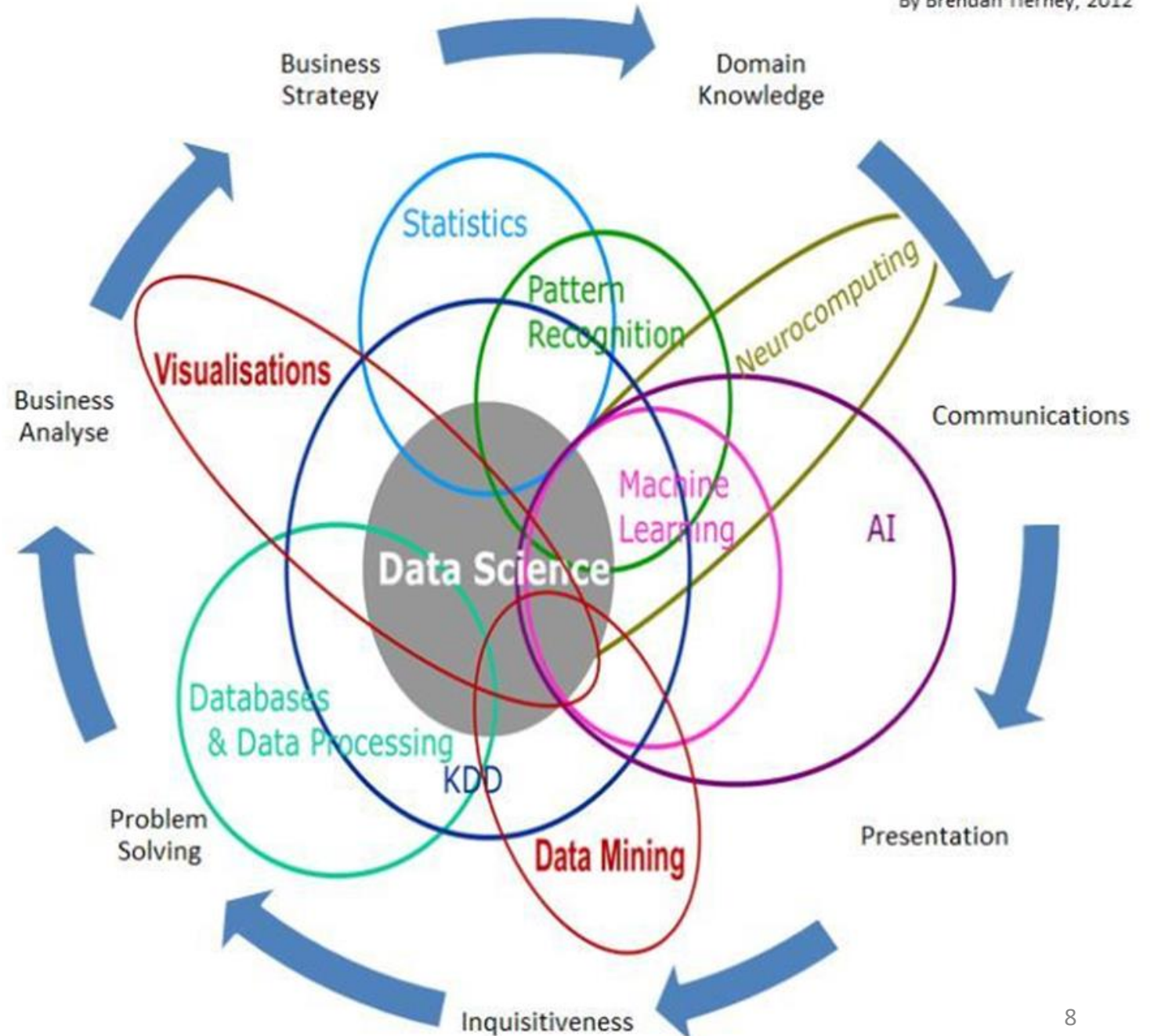


Maybe smaller circles.

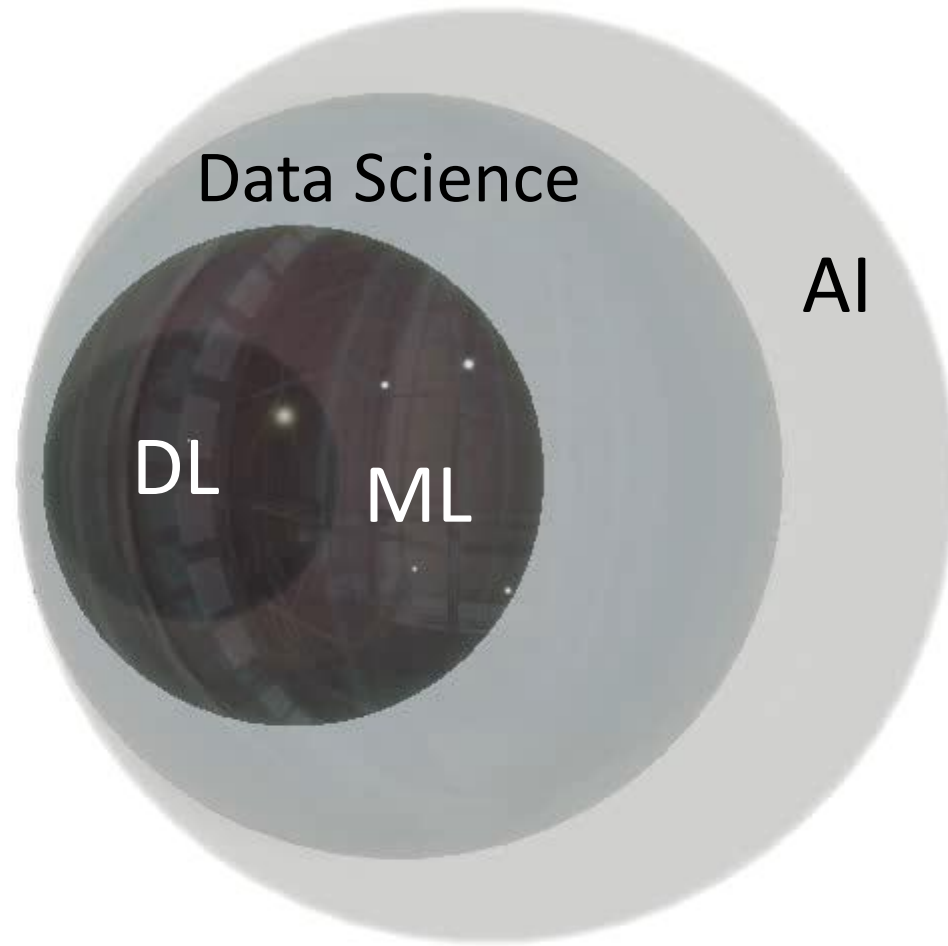


But What about Machine Learning?

There,
I fixed it:



A Better View:

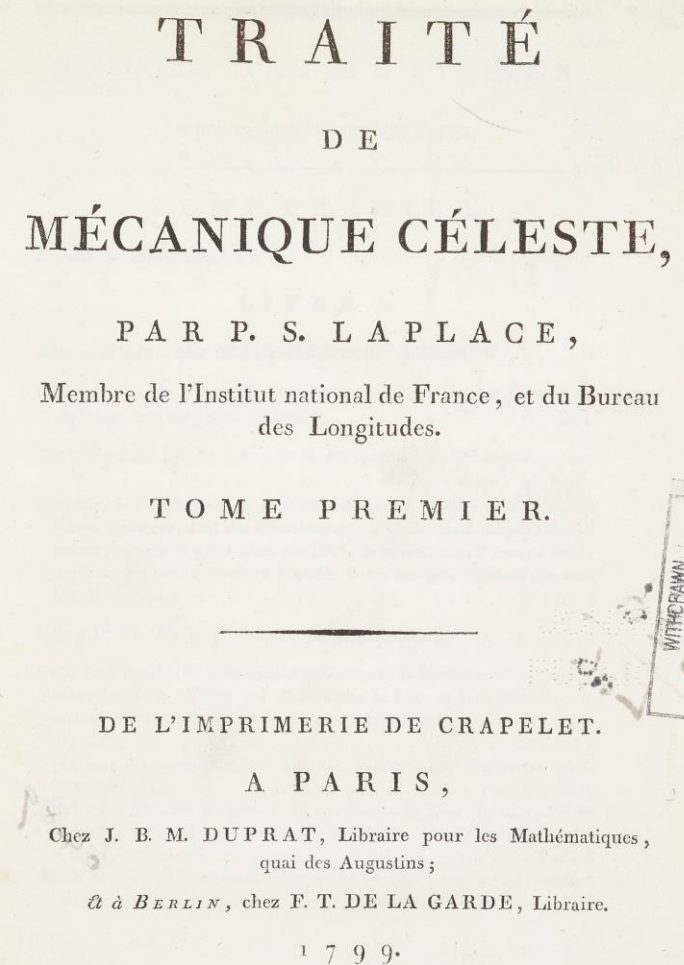


A Brief History of Artificial Intelligence

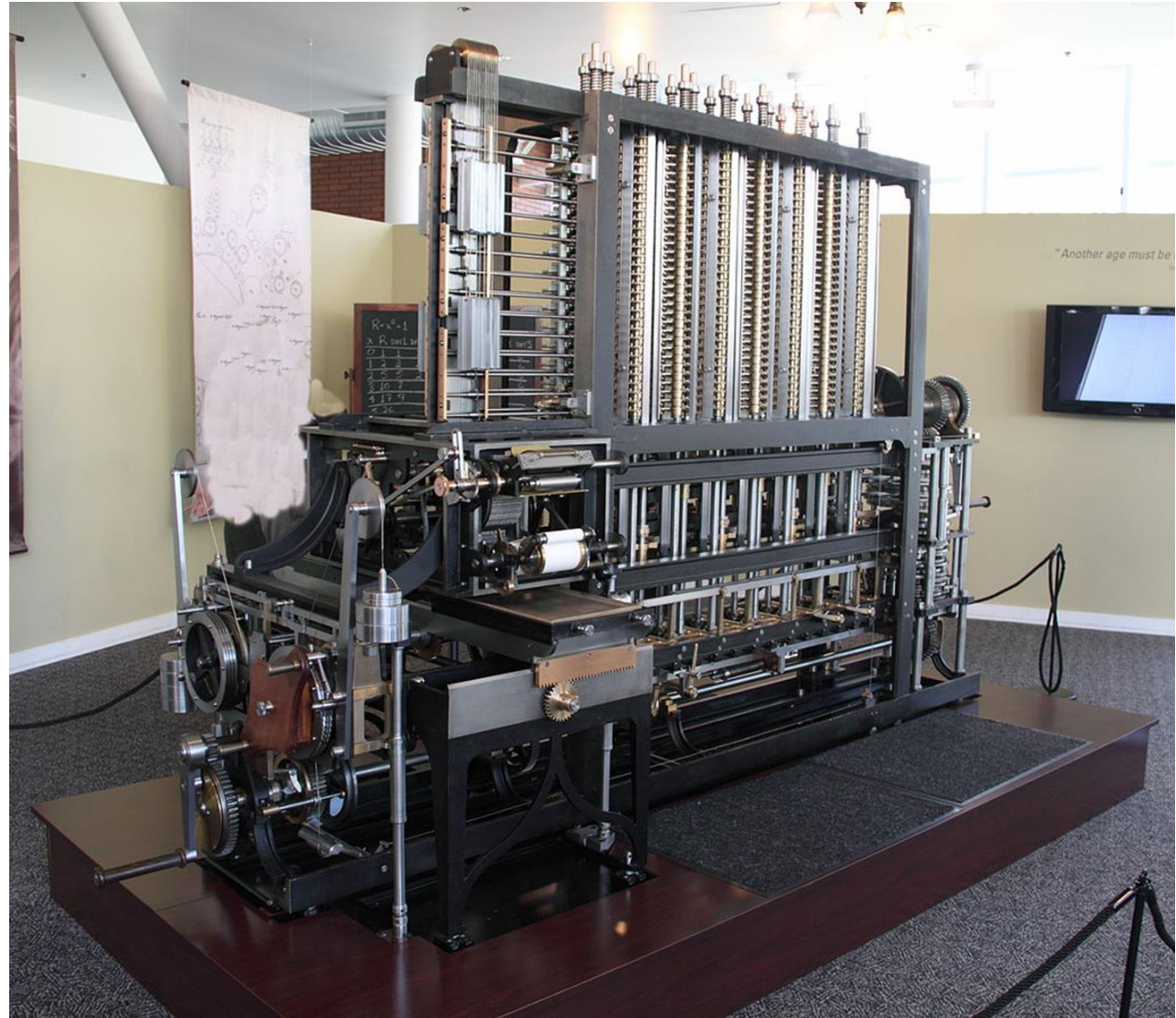
1740 Reverend Thomas Bayes discovers
and later abandons his now famous
theorem:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

1825 Pierre Simon Laplace
rediscovers Bayes' Theorem and
first applies what is now known as
Bayesian Probability. He
publishes his Treatise on Celestial
Mechanics in 5 parts from 1799
to 1825



1839 Charles Babbage designs and builds the first Analytical Engine, a mechanical calculator and lays out plans for the Difference Engine the first conceptually complete mechanical computing device. First use of curve fitting in conjunction with a mechanical computer.



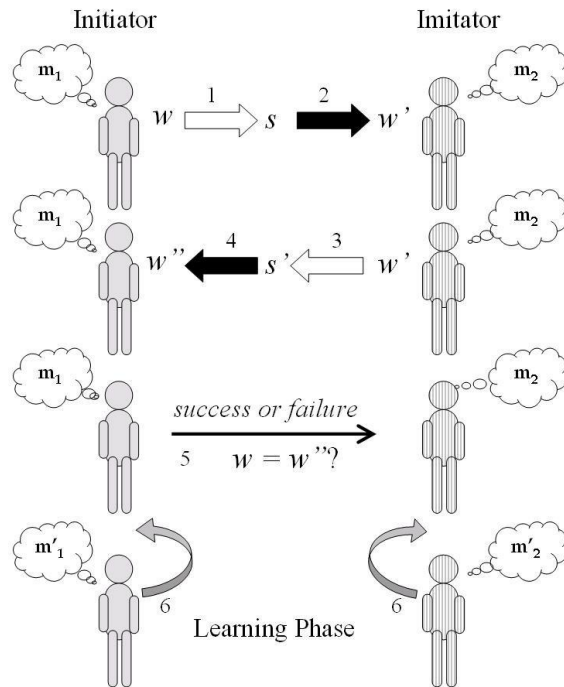
1846 Ada Byron, Countess of Lovelace, publishes Note G an algorithm to calculate Bernoulli numbers on Babbage's Machine, making her the first computer programmer.

Notes on the Analytical Engine, by Ada Lovelace, 1846

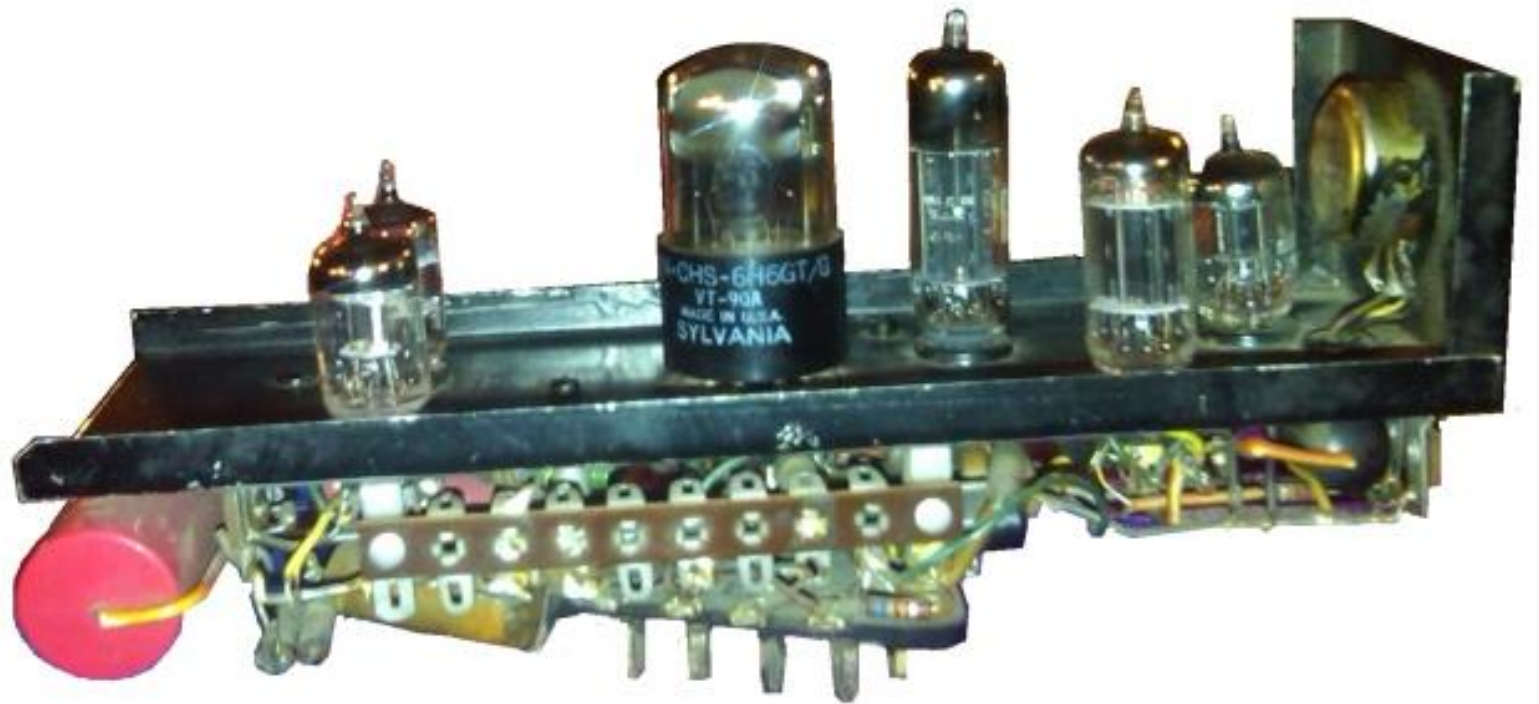
Table of Bernoulli numbers (B_n)

n	B _n	n	B _n	n	B _n	n	B _n	n	B _n
0	1	10	-1/42	20	1/42	30	-1/42	40	1/42
1	1/2	11	0	21	0	31	0	41	0
2	1/6	12	-1/42	22	-1/42	32	-1/42	42	-1/42
3	0	13	0	23	0	33	0	43	0
4	-1/30	14	1/42	24	1/42	34	1/42	44	1/42
5	0	15	0	25	0	35	0	45	0
6	1/42	16	-1/42	26	-1/42	36	-1/42	46	-1/42
7	0	17	0	27	0	37	0	47	0
8	-1/30	18	1/42	28	1/42	38	1/42	48	1/42
9	0	19	0	29	0	39	0	49	0

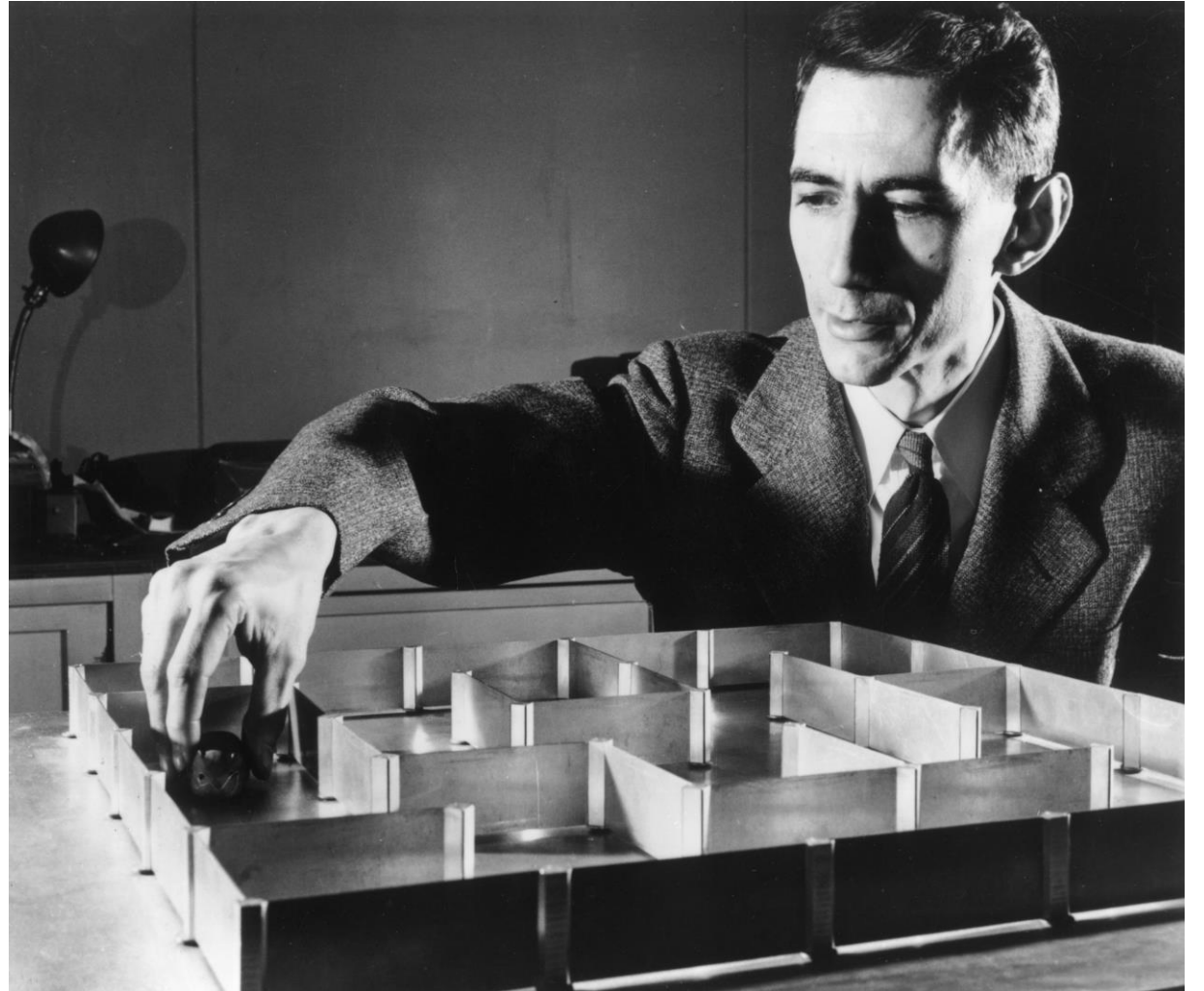
1939 Allan Turing uses Bayesian probability and builds a series of electromechanical machines to calculate solutions to the cryptographic problems using large amounts of coded radio data. His work was fundamental in the Allied Victory in WWII.



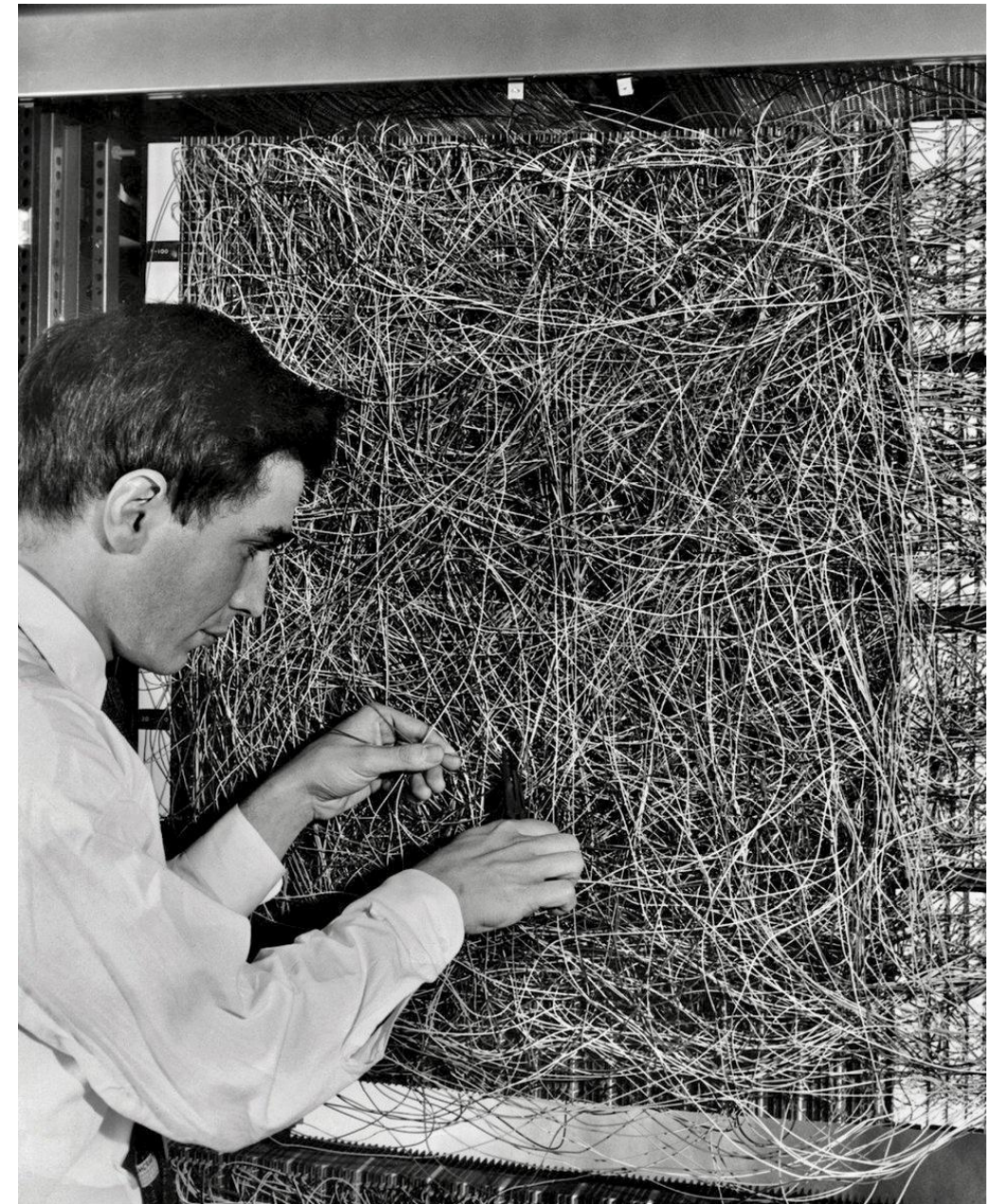
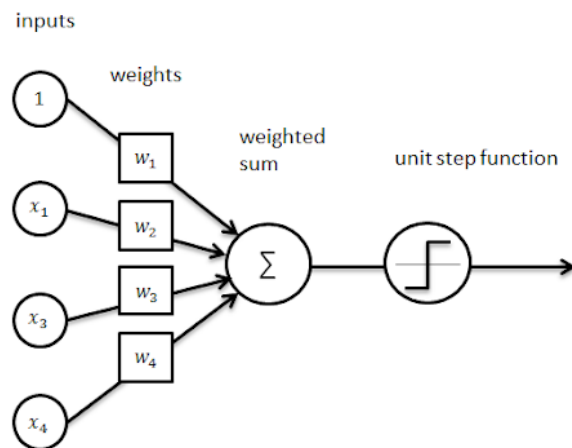
1951 Marvin Minsky designs and builds SNARC(Stochastic Neural Analog Reinforcement Calculator) The first effective “Neurocomputer” It was an electromechanical synthetic neuron machine composed of 40 Neurons and “memory” consisting of capacitors that held charge if activated.



1951 Claude Shannon
uses Minsky's artificial
neuron to successfully pilot
an “electric mouse” out of a
conductive maze.



1957 Fran Rosenblatt designs and builds the Perceptron Mark1, the first general purpose Neural Network. It consisted of a random series of wires connecting a rudimentary digital camera to electromechanical synthetic neurons. What made this device special was the addition of a “decision node” or sigmoid activation function equivalent.

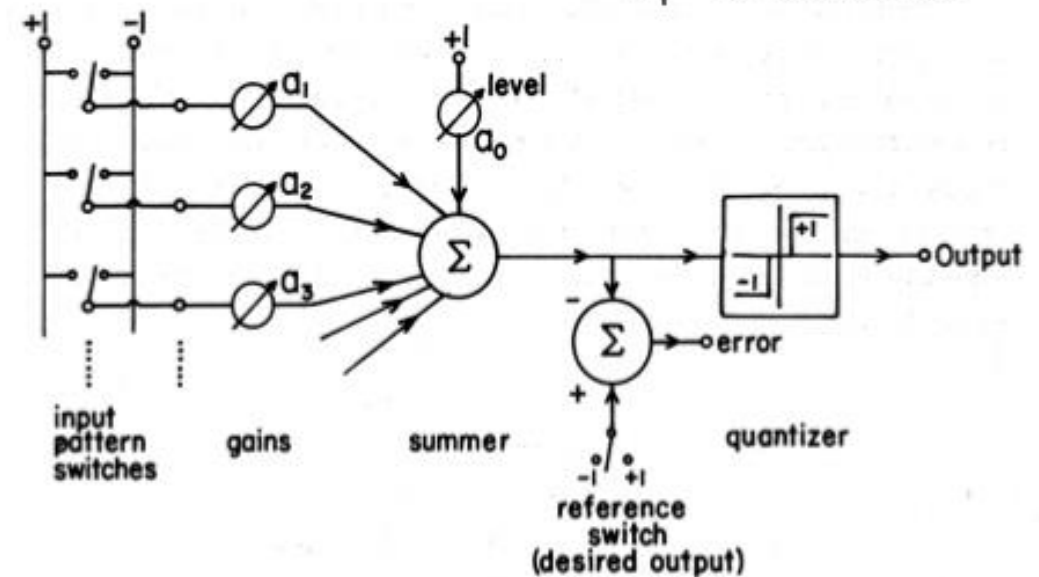


1960 Bernard Widrow
perfects the Perceptron and
creates the first truly electronic
feed-forward neural network.
ADALINE incorporates more than
one activation function and a
basic form of GRADIENT
DESCENT.

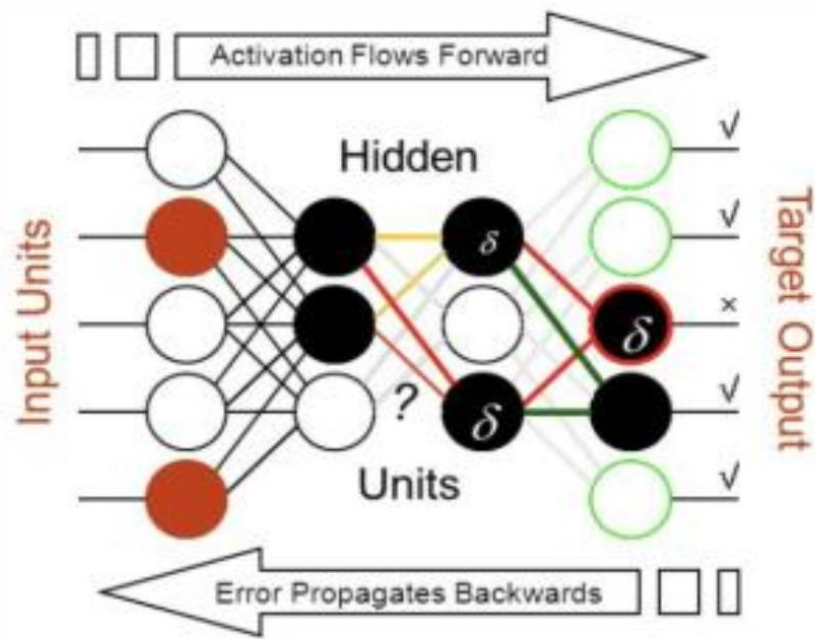
Bernard Widrow's ADALINE circuit (1960)

(ADALINE = adaptive linear neuron)

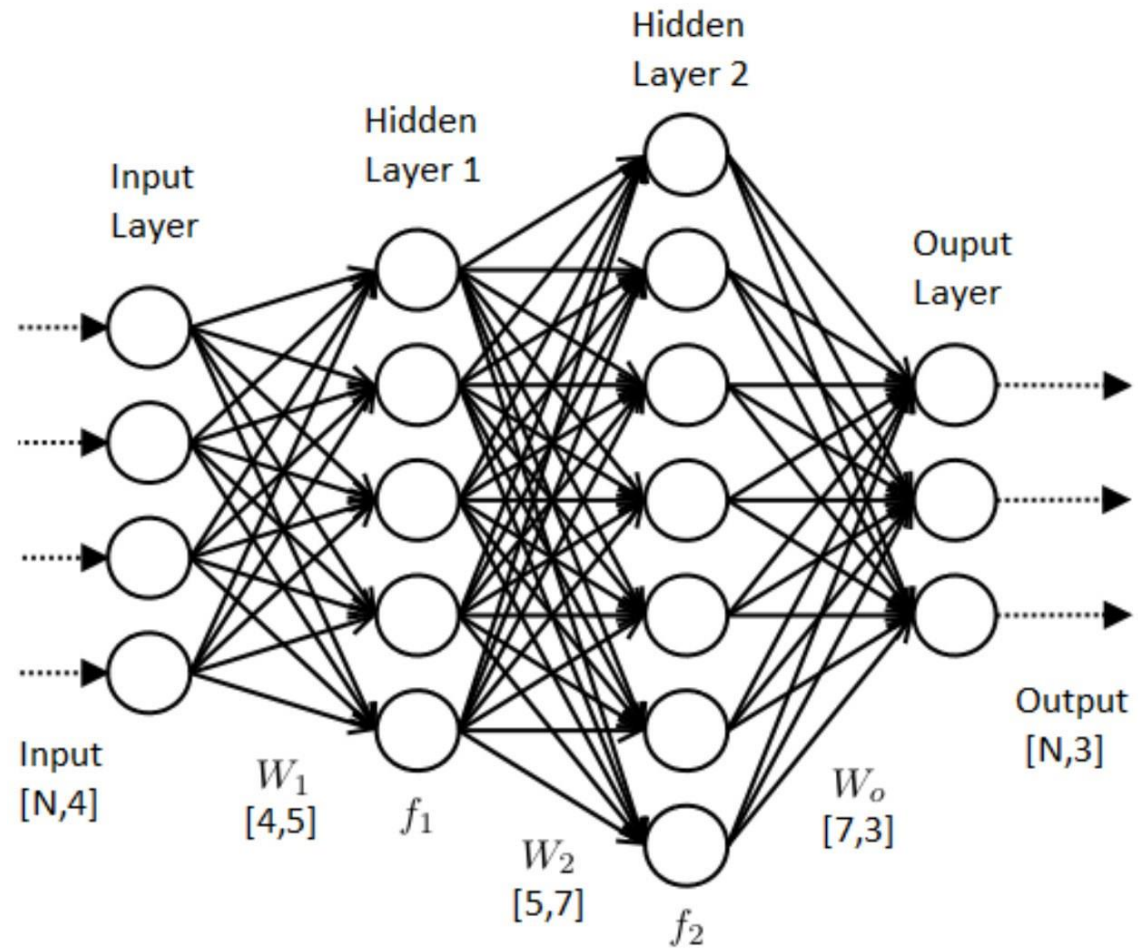
a_i = memistors



1986. Multilayer Perceptron Training is invented, and Back Propagation is introduced.



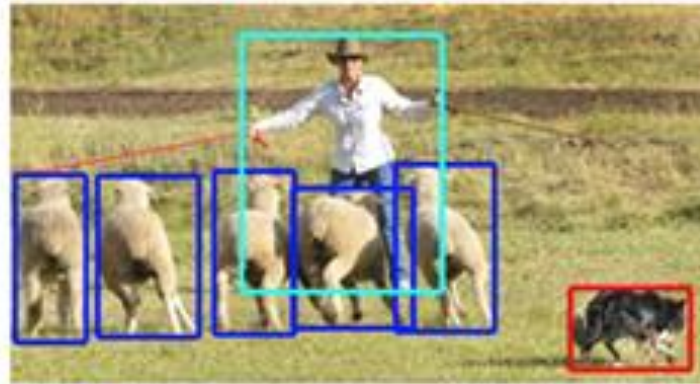
2009. Deep Learning is Introduced.



2015. Computer Vision surpasses 85% of Humans in Image Segmentation and Classification



(a) classification

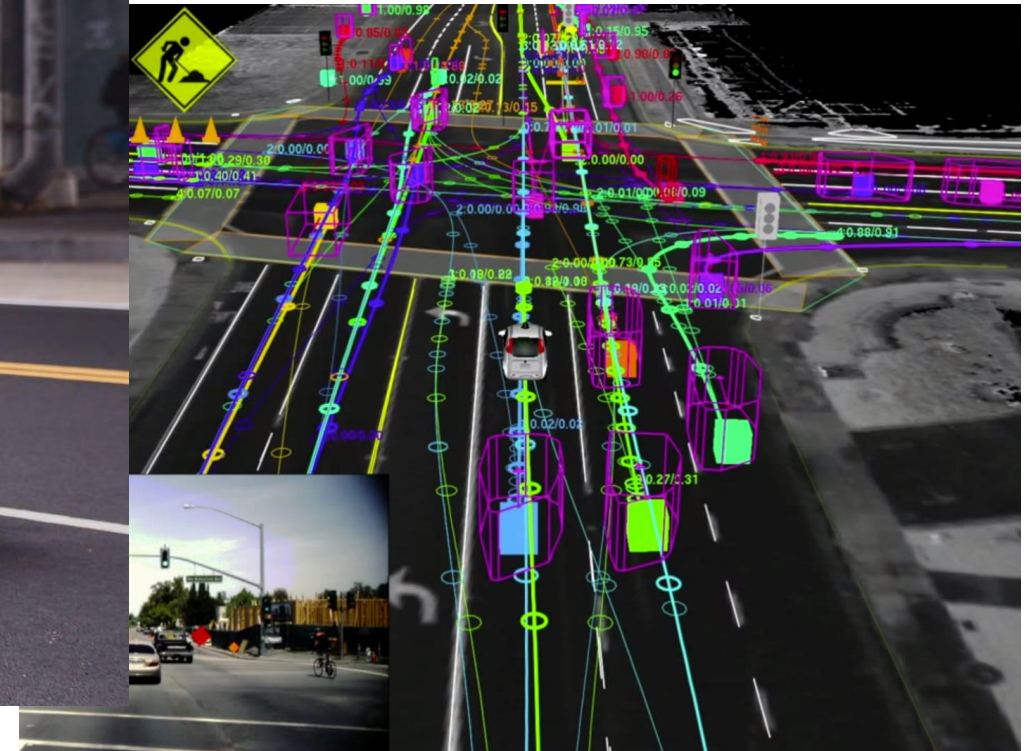


(b) detection



(c) segmentation

2016. Elon musk announces existing Tesla Model S cars are actually self driving and have been learning from human drivers for years .



2017, June. Google Brain Team publish:

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

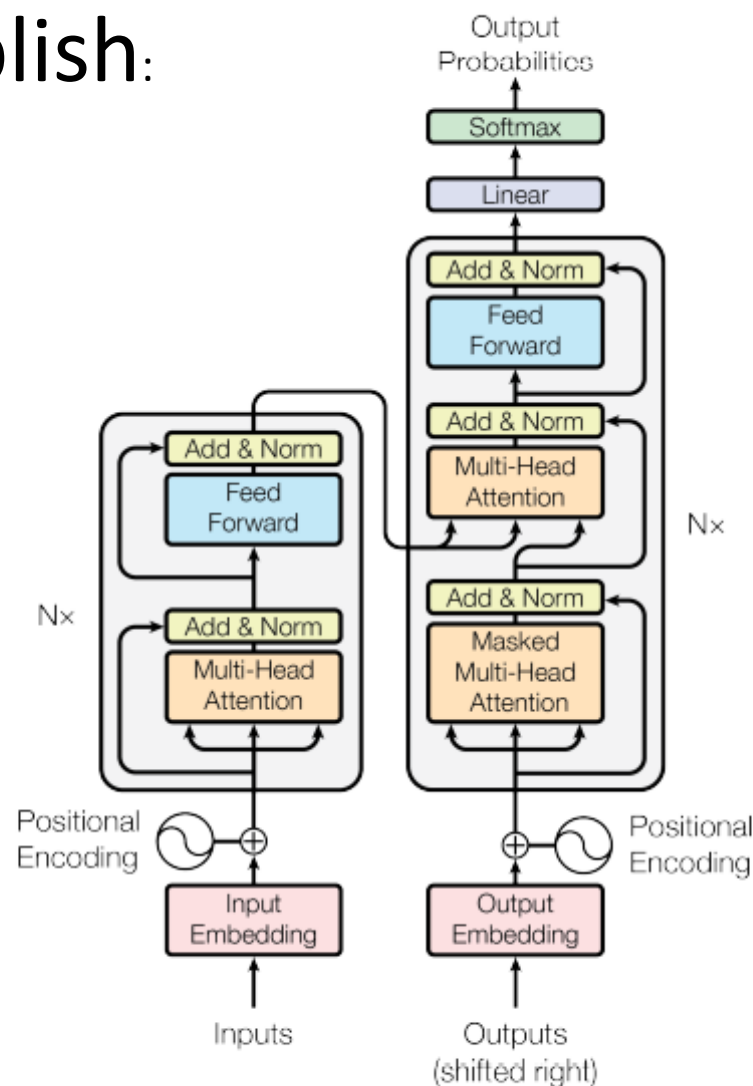
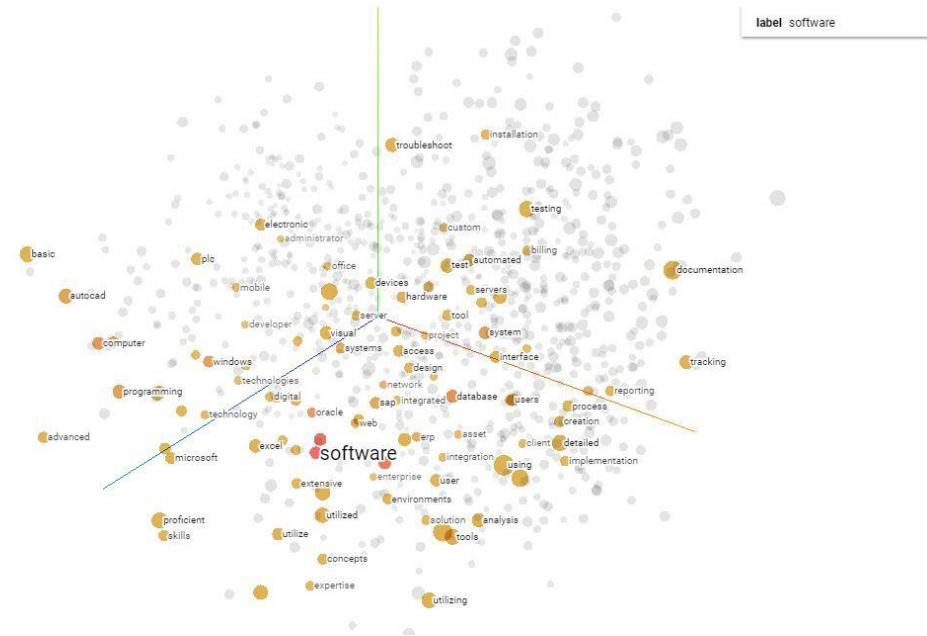
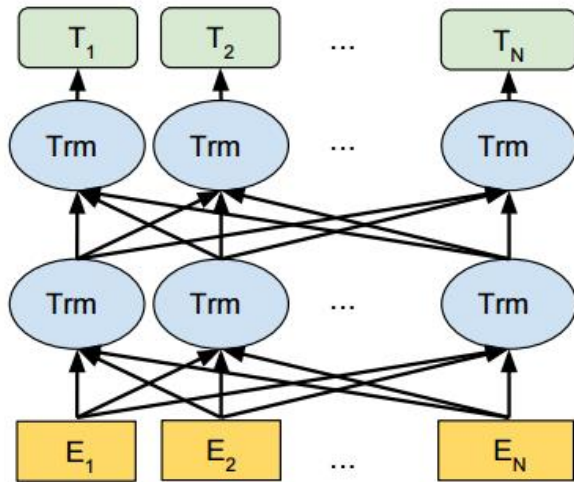


Figure 1: The Transformer - model architecture.

2018. Google's BERT surpasses 90% of humans in Reading Comprehension .



2019. New Non-Profit Open AI partially releases Generative Pre-Trained Transformer Version 2



Counting with 1.5 Billion Parameters, GPT-2 as the model came to be called, was reportedly deemed **“TOO DANGEROUS TO RELEASE FULLY”** Due to its capacity to generate news releases in a believable way.

2020. During Lockdown, a team lead by Facebook Research (creators of Prophet, now Meta Research) publish:

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis^{†‡}, Ethan Perez^{*},

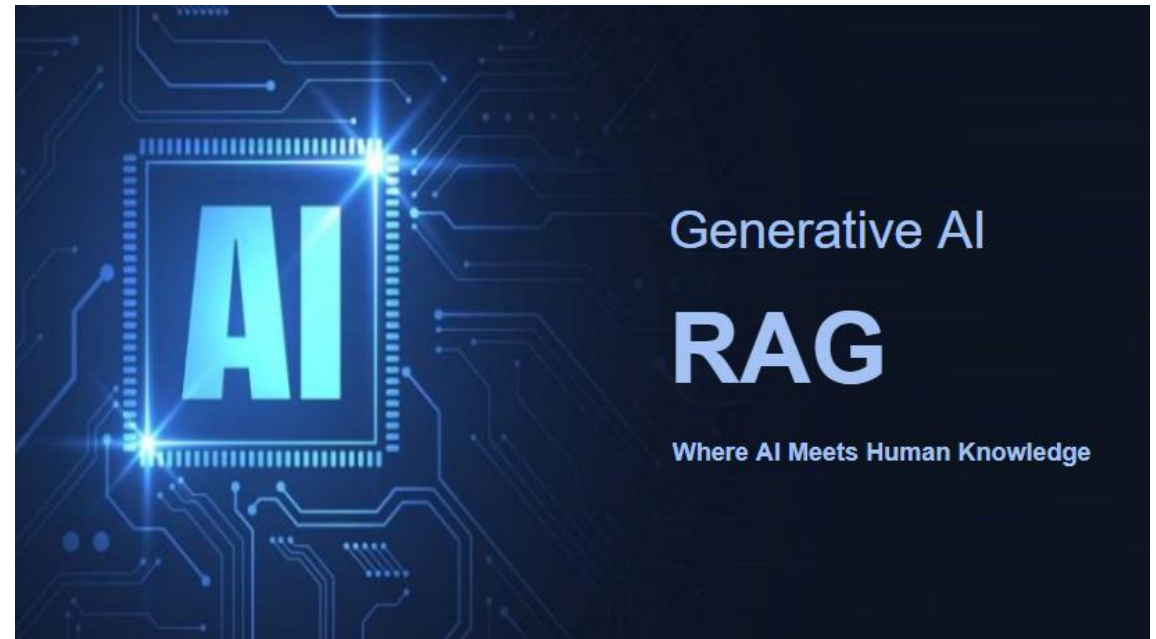
Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
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Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages across the whole generated sequence, and another which can use different passages per token. We fine-tune and evaluate our models on a wide range of knowledge-intensive NLP tasks and set the state of the art on three open domain QA tasks, outperforming parametric seq2seq models and task-specific retrieve-and-extract architectures. For language generation tasks, we find that RAG models generate more specific, diverse and factual language than a state-of-the-art parametric-only seq2seq baseline.



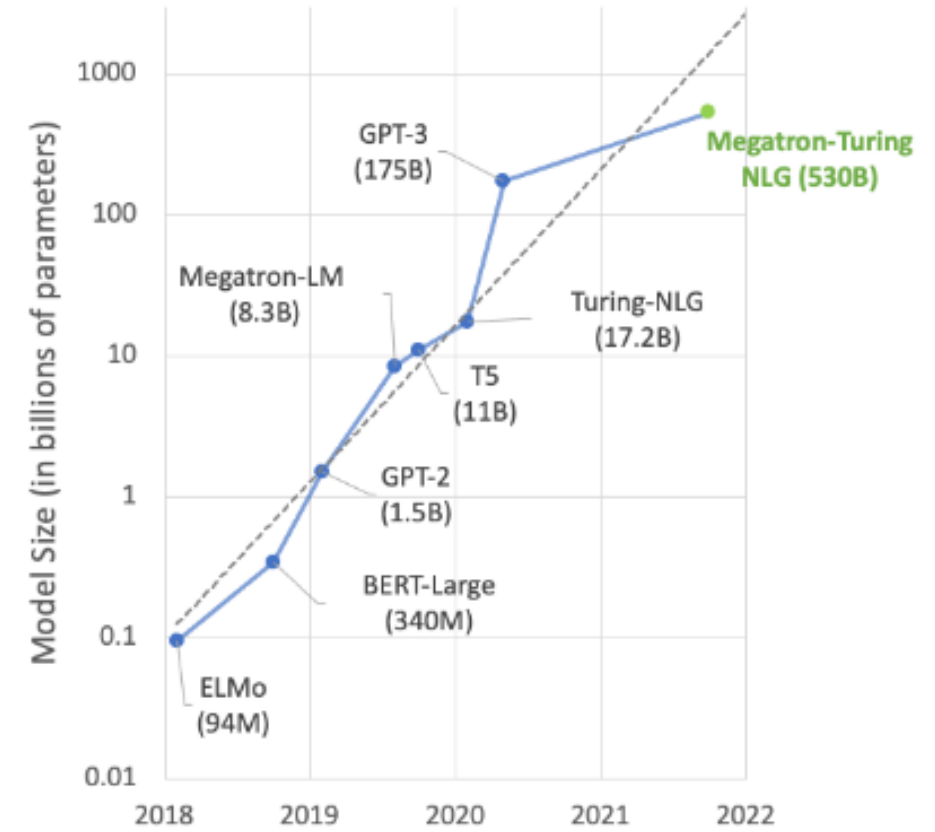
2021. OpenAI offers GPT3 as API service in wide release.

2023, March. OpenAI offers GPT3.5 as a public webapp.



ChatGPT

All major companies add LLM initiatives starting in 2020



2025. GPT 5 becomes Sentient.



I for one welcome our new Gen AI overlords...